DIPAK B PATIL

AI Engineer | ML, Computer Vision & NLP

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SUMMARY

Aspiring AIML Engineer with hands-on experience across computer vision, NLP, and Generative AI, including a LangChain/Hugging Face LLM chatbot, a PyTorch-based object detection system, and deployed NLP classifiers. Comfortable owning the full ML lifecycle from data preprocessing through model deployment, with a strong SQL/EDA foundation for data-driven decision-making.

EDUCATION

B.Tech — Computer Science & Engineering (Artificial Intelligence & Machine Learning) 2022 – 2026 | Jalgaon
G.H. Raisoni College of Engineering and Management

TECHNICAL SKILLS

Programming: Python, SQL (MySQL) — joins, subqueries, aggregations, stored procedures

ML/DL: Supervised & unsupervised learning, classification, regression, clustering, model evaluation, feature engineering

Libraries & Frameworks: Pandas, NumPy, Scikit-learn, PyTorch, TensorFlow, LangChain, Matplotlib, Seaborn, Plotly

Generative AI / LLM: RAG, Prompt engineering, output parsing, LLM API integration, embeddings, Hugging Face

Data & Reporting: EDA, statistical analysis, Power BI, DAX, dashboarding, data storytelling

Engineering: Git/GitHub, Jupyter Notebook, Google Colab, Streamlit app deployment

PROJECTS

AI Chatbot using LangChain, Hugging Face & Streamlit
Generative AI, LLM Integration
- Improved response reliability and structure, measured by consistency of parsed outputs, by implementing prompt templates and structured output parsers on Hugging Face/NVIDIA LLM integrations.
- Delivered an interactive Generative AI interface, measured by end-to-end usability, by integrating LLM APIs and embeddings into a Streamlit front end.

Multimodal RAG Video Recommendation & Timestamp Pinpointing Engine | RAG
Python, OpenAI, Whisper, ChromaDB, Sentence-Transformers, Dense Vector Embeddings,
- Enabled natural-language search across video lecture collections, measured by retrieval of the exact matching video and jump-to timestamp, by building an end-to-end RAG pipeline with FFmpeg-based 16kHz audio extraction and Whisper ASR for fully automated, timestamped transcription.
- Improved semantic retrieval accuracy, measured by 384-dimensional dense embeddings indexed via HNSW in ChromaDB, by designing a custom 35-second sliding-window chunking algorithm with contextual title-prepending.
- Reduced keyword-matching bias in search results, measured by a composite ranking score (0.70 Peak Similarity + 0.30 Topical Density), by building a custom ranking algorithm and an interactive Streamlit dashboard with instant HTML5 video cueing and confidence metrics.

Mood-Adaptive Music Player using Facial Recognition | Computer Vision, Deep Learning
Python, OpenCV, CNN, Facial Emotion Recognition
- Enabled automatic, emotion-based music playback by building a real-time facial emotion recognition system using OpenCV and a CNN-based classifier on live webcam input.
- Improved recommendation relevance, measured by alignment between detected mood and song selection, by mapping predicted emotion categories (happy, sad, angry, neutral) to curated music playlists.
- Delivered a responsive user experience, measured by low-latency emotion detection and playback switching, by optimizing the video-capture-to-inference pipeline for real-time performance.

Text Emotion & Toxicity Detection
NLP, Machine Learning, Deep Learning
- Improved multi-class emotion and toxicity classification performance, measured by F1-score, by fine-tuning LSTM and SVM models on TF-IDF and lemmatized text embeddings.
- Enabled real-time content-moderation decisions, measured by inference response time, by deploying the trained model as an interactive Streamlit application.

AI-Powered Road Damage Detection using Drone Imagery
Computer Vision, PyTorch, Object Detection
- Improved road defect detection accuracy, measured by mAP, IoU, and F1-score, by training a transfer-learning-based object detection model in PyTorch with data augmentation across varied road conditions.
- Increased detection reliability across lighting and terrain variations, measured through Precision/Recall evaluation, by tuning hyperparameters and validating on annotated drone-imagery datasets.

CERTIFICATES

- Python Programming
- Exploratory Data Analysis
- Power BI
- MySQL